Spatial embedding of $3+1$ causal sets: low-count spacelike pairs and a likelihood fit to interval counts

Roberto Bada[1, 2, *]

[1] *SapereIA, Barcelona, Spain*
[2] *Institute of Cosmos Sciences, University of Barcelona (ICCUB), Barcelona, Spain*

(Dated: October 30, 2026)

Recovering spatial distances from the causal order alone is a central open problem in the kinematics of causal sets. We study it on the benchmark of Johnston's recent embedding of $3+1$ causal sets into Minkowski spacetime [Phys. Rev. D **111**, 106020 (2025)]. After reproducing his results, we show with oracle substitutions that the residual error of the method comes entirely from spacelike pairs, and almost entirely (94–97%) from pairs whose shared causal regions contain fewer than ten elements. These pairs are a third to a half of all spacelike pairs at 1500–3000 elements, concentrate at the waist of the causal diamond, and become rarer only slowly with density; distances inverted from such counts also improve slowly, consistent with a scaling argument. We therefore fit the spatial coordinates directly to the interval counts of all related pairs, each with its Poisson weight (the likelihood of Casasola), starting from a map built from causal-overlap counts. On 20 fresh sprinklings with 3000–15000 elements the fit improves on Johnston's embedding in every universe and removes on average 79–83% of his residual at each size, and at $n = 3000$ halves the median error of individual spatial distances (from 0.22 to 0.11 discreteness lengths) and cuts the 90th percentile by a factor of three. In a cylindrical region at equal density the gain is 58–65%; intervals truncated at the timelike boundary are sufficient to account for the difference from the diamond, and the gain requires a minimum density.

I. INTRODUCTION

In causal set theory spacetime is modeled as a locally finite partially ordered set, whose elements are spacetime events and whose order is the causal order [1, 2]. For a causal set obtained by a Poisson sprinkling into a Lorentzian manifold, the number of elements in a region estimates its spacetime volume, and the order encodes the light cones; the kinematic programme is to recover the remaining geometry from these two ingredients. Timelike quantities are comparatively accessible: the longest chain between two related elements estimates their proper time [3], and interval counts give dimension and curvature [4–6]. Spatial distances are harder, because two spacelike elements bound no interval.

Several constructions address this. Brightwell and Gregory and, in more detail, Rideout and Wallden defined spatial distance through intervals between common past and future elements [3, 7]. Boguñá and Krioukov measured it through the overlap of the causal pasts of two elements as seen from a common past element [8], and Hu, Meyer and Meyer gave an exact $1+1$ reconstruction [9]. Embeddings of whole causal sets into Minkowski spacetime were constructed by Henson [10], by Clough and Evans through Lorentzian multidimensional scaling [11], and by Johnston, first through factorization of matrices of inner products [12] and recently through a simpler method [13] that is the current benchmark for $3+1$ sprinklings into a causal diamond. In that method the spatial separation of a spacelike pair is the shortest detour through an element related to both.

In this paper we ask where the remaining error of Johnston's embedding comes from, and how to remove it. Our results are the following.

1. We reproduce his benchmark and show by oracle substitution that the residual error is entirely due to spacelike pairs, and almost entirely due to those whose shared causal regions contain fewer than ten elements (Secs. III and IV).

2. With a causal-overlap distance estimator built on Boguñá and Krioukov's construction, extended to reference elements at unequal depth, we show that these low-count pairs also limit the classical-scaling embedding of inverted distances, and give a scaling argument for why their errors decrease only slowly with density (Sec. IV).

3. We fit the spatial coordinates directly to the interval counts of all related pairs, with their Poisson noise. The likelihood is Casasola's [14]; our contributions are an intrinsic starting map, the use of all related pairs, and a controlled comparison with Ref. [13] up to $n = 15000$, where the fit removes about 80% of the residual error (Sec. V).

4. We repeat the comparison in a region without a waist at equal density, and identify the timelike boundary as the source of the reduced gain (Sec. VI).

The cost of the methods is compared in Sec. VII, and Sec. VIII discusses novelty, limitations and extensions. Other approaches that we tested are summarized in Appendix A. A companion paper develops exact counting tools in $1+1$ dimensions [15].

* roberto@sapereia.com

II. SETUP AND BENCHMARK

Following Ref. [13], a Poisson number of points with mean n is sprinkled with density ρ into a causal diamond of height 1 in $3+1$ Minkowski spacetime (waist radius $\frac{1}{2}$, volume $\pi/24$), and the bottom and top tips v_1, v_n are added as two further elements. The order is induced by the light cones; the coordinates are then discarded and used only for scoring. We write $\ell = \rho^{-1/4}$ for the discreteness length, so that a region of volume ℓ^4 contains one element on average; at $n = 3000$ the diamond height is $\approx 12\,\ell$. For a related pair $a \prec b$, N_{ab} denotes the number of elements strictly between them; in a flat interval of proper time τ its mean is $\rho\,\pi\tau^4/24$.

A method returns coordinates from the order alone. As in Table V of Ref. [13], the score is the Pearson correlation $\mathcal{C}$ between embedded and true spatial distances over all pairs. We report the residual $1-\mathcal{C}$ and, for a new method, the fraction of Johnston's residual that it removes, $(\mathcal{C}-\mathcal{C}_{\mathrm{J}})/(1-\mathcal{C}_{\mathrm{J}})$, always on the same universes. Because a correlation close to one is dominated by distant pairs and insensitive to an overall rescaling, we also report per-pair distance errors and position errors in units of ℓ (Sec. V).

Three sets of sprinklings are used, and every number below states which: (a) the *large-n set*, 21 universes from $n = 1000$ to 15000 with independent seeds, used for Table II and Fig. 1; (b) *diagnostic sets* at $n = 1500$ (three universes) and $n = 3000$ (two universes), used for the low-count oracles of Sec. IV; (b′) *check sets* of the same sizes but different seeds, used for checks (ii) and (iv) of Sec. V; (c) the *shape comparison* of Sec. VI, four universes per shape and size. Sets (a), (b) and (b′) use Johnston's time coordinate, set (c) the true one. The reproduction and first oracle of Sec. III, the per-pair comparison of Sec. IV at $n = 1000$, and check (i) use separate development sprinklings. Quoted uncertainties are standard errors over universes.

III. JOHNSTON'S EMBEDDING

Johnston assigns each element a time coordinate from volumes and the proper time of related pairs from interval counts, so that the spatial separation of a related pair is $R = \sqrt{\Delta t^2 - \tau^2}$. For a spacelike pair he takes the shortest detour through an element z related to both,

$$r(x,y) = \min_z \big[R(x,z) + R(z,y)\big], \tag{1}$$

an upper bound by the triangle inequality, computed as a min-plus matrix product. Classical multidimensional scaling of r then gives three spatial coordinates.

Our implementation reproduces his Table V, $\mathcal{C} = 0.786,\ 0.887,\ 0.923$ at $n = 500,\ 1000,\ 1500$ (ten sprinklings each) against his 0.789, 0.887, 0.928, and his Table III, provided the proper time is computed from the elements strictly between the two endpoints; including the endpoints gives $\mathcal{C} = 0.72$ at $n = 1000$, as also noted by Pellegrin [16]. For the correlation of squared Minkowski intervals (his Table IV) we obtain 0.82, 0.90, 0.93 against his 0.958, 0.977, 0.985, also when the true time coordinate is used; we suspect that a different statistic is reported there, and use the spatial correlation throughout.

On development sprinklings, replacing the spacelike values r by the true spatial distances, and keeping every other step, raises $\mathcal{C}$ from 0.903 to 0.9999 at $n = 1000$ (from 0.915 to 0.9999 at $n = 1500$). Replacing instead the related-pair separations by their true values leaves it essentially unchanged ($0.903 \to 0.906$). The residual error of the method therefore resides in the spacelike pairs. Pellegrin, studying a link-restricted version of the detour, attributes much of its error to noisy related-pair distances of near-collinear pairs [16]; for Eq. (1) itself, correcting the related pairs does not help.

IV. OVERLAP DISTANCES AND LOW-COUNT PAIRS

Estimator. Let a, b be spacelike and c an element in their common past. In flat spacetime the volumes of $I(c,a)$, $I(c,b)$ and of their overlap C depend only on the proper times $\tau_{ca} \geq \tau_{cb}$ and on the spatial distance d. The overlap in units of the smaller interval,

$$w = \frac{V_C}{V_b} = w\!\left(\frac{\tau_{cb}}{\tau_{ca}}, \frac{d}{\tau_{ca}}\right) \in [0,1], \tag{2}$$

depends on two ratios only. We tabulate it once by Monte Carlo integration ($\tau_{cb}/\tau_{ca} \in [0.05, 1]$, $d/\tau_{ca} \leq 8$, common random numbers) and invert it in d, reading both proper times from interval counts through $V = \pi\tau^4/24$. On the causal set we use $w = (N_C + \frac{1}{2})/(N_b + 1)$, which keeps the inversion finite at $N_C = 0$, and cap the result at the detour of Eq. (1), an upper bound for exact separations. For the benchmark we take $c = v_1$, so that $d(a,b)$ requires only the sizes of the two pasts and of their intersection, and we use the future side ($c = v_n$) when it contains more elements. These counts for all pairs follow from two matrix products. The same three counts define Müller's past-cone metric [17], without an inversion to a distance.

This is the construction of Boguñá and Krioukov [8], who introduced the overlap of two causal intervals with a common past element as a measure of spatial distance and proposed embedding the resulting distance matrix. They invert the overlap exactly, numerically and in any dimension, for c at equal proper time below a and b [their Eq. (22)], and on a causal set keep only reference elements at equal depth [their Eqs. (32)–(34)]. Our extension is the table for unequal depths, $\tau_{cb}/\tau_{ca} < 1$, which allows a single reference element for all pairs. The Euclidean analogue, the distance between two sensors from the numbers of shared and unshared neighbors, is known in sensor-network localization, both as a deterministic inversion [18] and as a maximum-likelihood estimate with a Cramér–Rao bound [19]. For late elements of large

TABLE I. Spacelike spatial distances by true separation, $n = 1500$ (all spacelike pairs of one universe): median signed bias and median absolute relative error of the overlap estimate and of Johnston's detour, Eq. (1).

true $\lvert\Delta\mathbf{x}\rvert$	pairs	overlap		detour	
		bias	error	bias	error
0–0.1	3 872	−0.05	0.16	+0.40	0.48
0.1–0.2	43 341	−0.16	0.21	−0.08	0.34
0.2–0.35	217 706	−0.17	0.18	−0.12	0.22
0.35–0.5	295 991	−0.14	0.15	−0.07	0.13
0.5–1	230 652	−0.11	0.12	+0.01	0.06

sprinklings ($n = 2\times10^4$–8×10^4), whose shared pasts are large, the estimator has a median relative error of 1–2% and a bias below 1% for $d \geq 4\ell$, and 6–8% below 2ℓ.

Per pair and as an embedding. On the benchmark most pairs are not late, and the counts are much smaller. Table I compares the two estimates over all spacelike pairs, by true separation. The detour is poor for close pairs, where no element lies between them, and accurate for distant pairs, where many elements lie near the straight line; the overlap estimate behaves in the opposite way and is biased low by 5–17%. Restricted to pairs with at least 20 elements in the shared region, its median per-pair error is lower than that of the detour (0.112 against 0.132 at $n = 1000$; 0.062 against 0.083 at $n = 3000$). As a diagnostic, we embed the overlap estimates directly by classical scaling: the result is much worse than Johnston's ($1-\mathcal{C} = 0.18$ against 0.049 at $n = 3000$; Fig. 1). Per-pair accuracy does not translate into an accurate embedding.

Low-count pairs. Let k be the number of elements in the shared region on the side used by the overlap estimator. Pairs with $k < 10$ are 50% of the spacelike pairs at $n = 1500$ and 31% at $n = 3000$; most of them straddle the waist of the diamond, where both the common past and the common future are small. A second oracle locates Johnston's error on them (diagnostic sets): replacing his detour by the true distance on the $k < 10$ pairs only removes 97% ($n = 1500$) and 94% ($n = 3000$) of his residual, while the truth on all other spacelike pairs removes 3% and 4% (97–99% against 2% with the threshold $k < 30$). The same holds for the overlap estimates on the same universes: with the truth on the $k < 10$ pairs, their direct embedding removes 96% ($n = 1500$) and 88% ($n = 3000$) of Johnston's residual, while the truth on all other pairs leaves it far worse than his. Classical scaling gives the noisy low-count distances the same weight as all others, whichever estimator supplies them.

Why higher density helps slowly. The distance is carried by the elements in the past of b that are not in the past of a. For small d this region is a shell of thickness $\sim d$ on a cone of height τ, containing $\kappa\rho\, d\, \tau^3$ elements with $\kappa = O(1)$. Its Poisson fluctuation gives, as an order-of-magnitude estimate,

$$\frac{\delta d}{d} \sim \frac{1}{\sqrt{\kappa\rho\, d\, \tau^3}}. \tag{3}$$

At fixed shape all lengths grow as $n^{1/4}$ in units of ℓ, so $d\tau^3 \propto n$ and $\delta d/d \propto n^{-1/2}$ for a typical pair; the measured median errors fall as $n^{-0.54}$ (overlap) and $n^{-0.42}$ (detour) between $n = 1000$ and 3000. The tail, however, is geometric: a pair on opposite sides of the waist has a small common past *and* a small common future, so its τ in Eq. (3) is set by its position rather than by the density. Between $n = 6000$ and 15000, the range least affected by small-n effects, the residual of the direct overlap embedding falls approximately as $n^{-0.4}$, against $n^{-0.85}$ for Johnston's (large-n set, Fig. 1); between $n = 3000$ and 6000 it still halves. This motivates using the counts themselves, with their noise, instead of distances inverted from them.

V. FITTING COORDINATES TO INTERVAL COUNTS

Likelihood. We keep Johnston's time coordinates t_i and treat the $3n$ spatial coordinates $\mathbf{x}_i$ as unknowns. For a related pair $a \prec b$, N_{ab} is Poisson-distributed with mean $\rho\,\pi\tau_{ab}^4/24$, where $\tau_{ab}^2 = (t_b - t_a)^2 - |\mathbf{x}_b - \mathbf{x}_a|^2$. We use this in Gaussian form for τ^2: with $\widehat{\tau^2}_{ab} = [(N_{ab} + \frac{1}{2})/(\rho\pi/24)]^{1/2}$ and $s_{ab} = \widehat{\tau^2}_{ab}/(2\sqrt{N_{ab}+1})$, which corresponds to a relative error $\delta N/2N$,

$$\ln L(\mathbf{x}) = -\frac{1}{2}\sum_{a\prec b}\left(\frac{\tau_{ab}^2(\mathbf{x}) - \widehat{\tau^2}_{ab}}{s_{ab}}\right)^2, \tag{4}$$

summed over *all* related pairs. Spacelike pairs do not enter. Each element is constrained by its $O(n)$ related partners, a pair with many elements between it tightly and one with few loosely; this is the weighting that classical scaling lacks. The Gaussian form is a convenience: the exact Poisson log-likelihood $\sum[N\ln\mu - \mu]$ gives the same maps (check (iv) below), provided it also encodes that the pair is related; otherwise a pair with $N = 0$ is best fitted as spacelike at no cost, and the fit drifts. The Gaussian form, whose target $\widehat{\tau^2}$ is always positive, needs no such term. We maximize Eq. (4) with the Adam optimizer and the analytic gradient (checked against finite differences), for 400 iterations with a step of 1% of the spatial spread, reduced by a factor 0.3 halfway; the tips are fixed at $\mathbf{x} = 0$. Each iteration costs $O(\#\text{related pairs}) = O(n^2)$.

Starting map. The fit requires a start that is roughly correct and is itself derived from the order. We embed the overlap estimates of Sec. IV by weighted stress majorization (SMACOF [20]; accuracy-weighted scaling is standard in sensor localization [21]), with weight $k/(k+100)$ for each spacelike pair and 1 for each related pair, so that low-count distances have little influence. This map alone

removes 23–47% of Johnston's residual, less at large n (Table II). The fitted result is unchanged when the fit is started from Johnston's own map instead: on the check sets the gain agrees within one percentage point (74–76% at $n = 1500$ and 3000).

Relation to previous work. Fitting causal-set coordinates to interval counts with a Poisson likelihood was proposed by Casasola [14], who used a pseudo-likelihood $\sum[N \ln \mu - \mu]$ over related pairs near the light cone ($\tau^2 < 6\ell^2$) in a $3+1$ diamond, with Langevin sampling started from the *true* positions displaced by about ℓ; an intrinsic start was left as an open problem, and no comparison with Ref. [13] was made. Maximum-likelihood coordinates are standard in other contexts: multidimensional scaling [22], latent-space network models [23], hyperbolic maps of complex networks, including a geometric first guess refined by likelihood as here [24, 25], and directed graphs in Minkowski space [26]. What we add is the intrinsic start, the use of all related pairs, and the large-n test on Johnston's benchmark.

Results. Table II and Fig. 1 show the fit on the large-n set, paired with Johnston's embedding on the same universes. The fit improves on his embedding in each of the 21 universes, and removes 79–83% of his residual at every size from $n = 3000$ to 15000 (67% in the single universe at $n = 1000$). The removed fraction shows no trend with n, so the gap does not close. At $n = 15000$ the spatial correlation is 0.9980 against 0.9886. The oracle value with exact spacelike distances, 0.9999, remains higher.

Per-pair and per-element errors. We recomputed the maps of the large-n set at $n = 1000$ (one universe), 3000 (two) and 6000 (four) on a laptop; the correlations reproduce Table II. In units of the discreteness length, the median absolute error of the embedded spatial distances is 0.29, 0.22, 0.17 for Johnston's embedding and 0.14, 0.11, 0.11 for the fit, and the 90th percentiles are 1.10, 0.80, 0.57 against 0.38, 0.28, 0.26; the 99th percentile at $n = 6000$ is 2.4 against 0.46. For positions, after removing the best orthogonal map, the median error per element at $n = 3000$ is 0.24–0.30 for Johnston's embedding and 0.16–0.17 for the fit, with 90th percentiles 0.57–0.75 and 0.29–0.32. The rms position error improves less ($0.67 \to 0.52$ at $n = 3000$, $0.56 \to 0.43$ at $n = 6000$), because it is dominated by a few elements: the tenth of the elements with the fewest related partners carries an rms of 1.4–1.6 in the fit, the other nine tenths 0.18–0.19. Neither a global rescaling nor an affine map removes this part, and fitting with the true time coordinate instead of Johnston's (whose time error is only $0.07\,\ell$ rms) changes it little ($0.49 \to 0.48$ and $0.55 \to 0.49$ in the two universes). In the single universe at $n = 1000$, whose Johnston correlation (0.857) is low for its size, the fit improves the distances but not the rms position ($0.88 \to 0.87$). Started from the true positions, the fit stays within 0.21 rms of them and reaches a higher log-likelihood than from the intrinsic start (by 105–109 over 4.7×10^5 related pairs at $n = 3000$, and by 228–549 over 1.8–1.9×10^6 at $n = 6000$), so the residual reflects a local optimum reached from the starting map in weakly constrained regions, not a floor of the likelihood.

TABLE II. Likelihood fit, Eq. (4), against Johnston's embedding on the large-n set, paired on the same universes. Gain: $\mathcal{C} - \mathcal{C}_\mathrm{J}$ with the standard error of the paired differences in parentheses (only indicative for two universes). Last two columns: fraction of Johnston's residual removed by the fit and by its starting map.

n	univ.	$\mathcal{C}_\mathrm{J}$	$\mathcal{C}$ (fit)	gain	fit	start
3000	2	0.9515	0.9901	0.0386(24)	80%	47%
6000	4	0.9752	0.9953	0.0201(8)	81%	38%
10000	8	0.9846	0.9968	0.0122(3)	79%	23%
15000	6	0.9886	0.9980	0.0094(2)	83%	26%

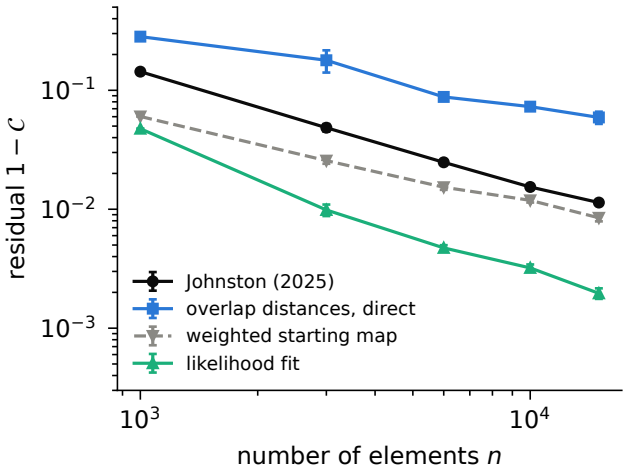


FIG. 1. Residual $1 - \mathcal{C}$ of the spatial-distance correlation on Johnston's benchmark (large-n set; $n = 10^3$–1.5×10^4 with 1, 2, 4, 8 and 6 universes): his embedding [13], the direct embedding of overlap distances (Sec. IV), the weighted starting map, and the likelihood fit, Eq. (4). Error bars: standard error over universes.

Causal fidelity. Spacelike pairs do not enter Eq. (4), so we checked how much of the order the fitted coordinates reproduce. At $n = 3000$ the embedded map misclassifies 0.65–0.69% of all pairs (related as spacelike or vice versa), against 1.6–2.1% for Johnston's embedding (0.48–0.49% against 1.21–1.29% at $n = 6000$, two universes).

Checks. (i) *Consistency.* Started from the true positions (with true times), the fit departs from the truth only by finite-n noise and approaches it as n grows: $\mathcal{C} = 0.966$, 0.991, 0.9965 at $n = 400$, 1000, 2000. (ii) *Start independence*, shown above for the check sets. (iii) *No tuning on the test universes.* The large-n runs used settings fixed beforehand on other universes, including the start's weight scale 100. (iv) *Form of the likelihood and convergence.* On the check sets we compared the Gaussian form with the exact Poisson log-likelihood plus a soft relation term $\sum \ln \mathrm{sig}(\tau^2/s_0)$, with sig the logistic

function and $s_0 = 0.1\,\ell^2$ (cf. the soft walls of Ref. [14]), each at 400 and 1000 iterations. All four variants remove 76–77% ($n = 1500$) and 74–75% ($n = 3000$) of Johnston's residual, with paired differences of at most 3×10^{-4} in $\mathcal{C}$. (These fractions differ from those of Table II because the universes differ.) Started from the truth, the exact form remains slightly closer ($\mathcal{C} = 0.973$, 0.993, 0.997 at $n = 400$, 1000, 2000). The log-likelihood still increases slowly after 400 iterations, but the map no longer changes.

VI. A REGION WITHOUT A WAIST

Johnston's error concentrates at the waist of the diamond. We therefore repeat the comparison in a cylinder ("slab") $t \in [0, 1]$, $|\mathbf{x}| \le \frac{1}{2}$, without tips. There his embedding is already accurate ($\mathcal{C} = 0.985$ at $n = 1500$, against 0.926 in the diamond). The likelihood needs no reference element and applies unchanged, with no element fixed (we remove the mean of the gradient). To isolate the spatial step, all methods in this section use the true time coordinate; Johnston's detours fall back to four legs when no two-leg detour exists, and the starting map uses for each pair one of 64 reference elements near the lower and upper boundaries.

We compare the shapes at equal density: slab $n = 6000$ and 12000 against diamond $n = 1500$ and 3000, four universes each. The fit removes $58 \pm 9\%$ and $65 \pm 6\%$ of Johnston's residual in the slab, against $75 \pm 5\%$ and $80 \pm 6\%$ in the diamond through the same pipeline. The difference arises at the timelike boundary. A diamond contains every interval between its elements (it is causally convex); a slab does not, and an interval that extends beyond the boundary contains too few elements, so that its pair appears closer to null than it is, as Pellegrin observed for slab-shaped subsets of a diamond [16]. In the slab, 45% of related pairs lose part of their interval and 14% lose more than a fifth, at every n. Dividing each count by the true fraction of its interval inside the slab (an oracle used for diagnosis only) raises the slab result to 80% and 86% (means over the same four universes), level with the diamond within the uncertainties quoted above: the boundary is sufficient to account for the difference. Its cost, about 20 percentage points, does not decrease with n, because the fraction of truncated intervals is fixed by the shape. On interior pairs (both elements in the top quartile of $\min(|\text{past}|, |\text{future}|)$) the fit does at least as well in the slab as in the diamond ($80 \pm 12\%$ and $86 \pm 2\%$, against $69 \pm 10\%$ and $71 \pm 14\%$).

Two caveats apply. These are fractions of a small residual: in the slab Johnston's $1 - \mathcal{C}$ is 0.0069 and 0.0047, about ten times smaller than in the diamond. And the gain requires density: in the slab it drops to $43 \pm 10\%$ at $n = 1500$ (three universes), and in a single universe at $n = 600$ the fit is worse than his embedding.

VII. COMPUTATIONAL COST

Johnston's min-plus step costs $O(n^3)$ operations; the overlap counts require two dense matrix products and a table look-up; the likelihood costs $O(n^2)$ per iteration at a fixed 400 iterations. We timed both on one universe per size, on an exclusive node with four BLAS threads (the min-plus step is single-threaded). Johnston's pipeline (matrix product, min-plus step, eigendecomposition) took 44, 525, 2002 and 7691 s at $n = 3000$, 6000, 10000, 15000; the likelihood pipeline (counting step, starting map, set-up, 400 iterations) took 307, 1328, 3647 and 8239 s. Over $n = 6000$–15000 the wall times scale as $n^{2.92}$ and $n^{1.99}$. The likelihood is slower at every size we ran, by a factor that falls from 7.0 to 1.07; extrapolating the two power laws, the times would cross near $n \approx 16000$. The claim of this paper concerns accuracy, not speed.

VIII. DISCUSSION

What is new. The overlap construction is that of Boguñá and Krioukov [8], the time coordinate and the benchmark are Johnston's [13], and the likelihood of related-pair counts is Casasola's [14]; maximum-likelihood coordinates are standard in other fields [22–26]. To our knowledge the following are new: the tabulated overlap inversion for unequal depths; the reproduction and oracle analysis of Ref. [13], with the localization of its error on low-count spacelike pairs; the scaling argument for why inverted distances improve slowly; the intrinsic starting map; and the large-n comparison, with the analysis of a second region. Our literature search (October 2026) covered the citing papers of Refs. [8, 11, 13, 14] and sources beyond arXiv and INSPIRE; at the time of writing Ref. [13] has one citing paper on INSPIRE, which reconstructs curvature from counts with local charts rather than distances [28].

Limitations. The likelihood assumes a flat interval volume; in curved spacetime volumes acquire corrections of order $R\tau^2$, which would bias the fit increasingly with interval size (the $1 + 1$ analogue is the depth drift of the companion paper [15]). The time coordinate is Johnston's and is not refitted. The benchmark is a flat diamond with 2–8 universes per size at large n; in the slab the boundary costs about 20 percentage points at every size, and the gain requires a minimum density.

Outlook. The oracle value ($\mathcal{C} = 0.9999$ against 0.998 at $n = 15000$) leaves room for improvement. Natural next steps are a joint fit of time and space coordinates; a model of the boundary of regions that are not causally convex [in $1 + 1$ dimensions timelike boundaries can be identified from the order alone [27], which might supply the fractions that our boundary correction took from the truth]; and a curved interval volume, perhaps via the normal-coordinate construction of Ref. [28], which would turn the fit into a curvature estimator.

DATA AVAILABILITY

The code that produces every number in this paper, and the research notebook in which the work was carried out, are openly available at https://github.com/presindev/causet-counting. The overlap table is in `sims/r11_4d_distance.py`; the reproduction of Ref. [13] and the first oracle in `sims/r12_johnston_compare.py`; the low-count oracles in `sims/r18_waist_oracle.py` and `sims/r30_johnston_lowk_oracle.py`; the starting map in `sims/r19_cluster_wfit.py`; the likelihood in `sims/r23_likelihood.py` and its large-n runs in `sims/r24_cluster_lik.py`; the per-pair errors in `sims/r29_perpair_error.py` the per-element analysis in `sims/r33_time_vs_position.py` and the likelihood comparison and causal fidelity in `sims/r34_lnl_causal_fidelity.py`; the slab in `sims/r22_slab.py` and `sims/r25_slab_lik.py`; and Fig. 1 by `sims/fig_3p1.py --prd`.

ACKNOWLEDGMENTS

This work was funded by SapereIA. Simulations were run on the Nyx cluster of the Institute of Cosmos Sciences of the University of Barcelona (ICCUB). An AI system (Claude Opus 5.5, Anthropic) was used for literature search, derivations, code and drafting, under the author's direction and checks; the procedure is described in Ref. [29]. The author takes full responsibility for the content.

Appendix A: Other approaches tested

Before the likelihood fit we tried to make inverted distances competitive, alone or combined with Johnston's detours. Each item was tested on the same universes as the method it is compared with; fractions are of Johnston's residual.

- *Hard switch*: the overlap estimate when it places the pair closer than $L = 0.25$, Johnston's detour otherwise. With L chosen on other universes, it improved on his embedding in every universe from $n = 500$ to 15000, removing a stable 17–20% of his residual. In the slab it gains nothing: its gain is located at the waist.

- *Weighted map as a final result*: its share fades with n (Table II) and depends on the weight scale K_0 in $k/(k + K_0)$ (23–36% at $n = 10^4$–1.5×10^4 for $K_0 = 100$–300). Retained only as the starting map.

- *Binomial term for spacelike overlap counts*, $N_C \,|\, N_b \sim \mathrm{Bin}(N_b, w)$ with w from the table of Sec. IV, added to Eq. (4). Alone it removes 14–25%; added to the fit it lowers the gain to 58–64% at every n of the large-n set. We have not isolated why (the bias of Table I, ratios of noisy counts, and pairs treated as independent are candidates).

- *Map from counts only* (SMACOF with distant pairs at weight 0.1, no detours): matched Johnston up to $n = 3000$, worse at larger n (-14% to -31%). Dropping the distant pairs instead of down-weighting them: $\mathcal{C}$ lower by 0.012 at $n = 3000$.

- *Laplacian eigenmaps*, suggested in Ref. [8]: limited to $\mathcal{C} \approx 0.93$ even with true distances, because the eigenfunctions of a ball are not linear coordinates.

- *Local trilateration charts* following Ref. [28]: exact on true intervals, but 30–42% per-pair error from counts at $n \leq 3000$ (ill-conditioned Gram matrices).

- *Detour on the low-count pairs only*, overlap estimates elsewhere: no better than Johnston's embedding.

- *Averaging* the overlap estimate with the detour: worse than the detour alone ($\Delta\mathcal{C} = -0.07$ to -0.006).

- *Several reference elements* instead of v_1, averaged: no gain, since deep references see the same region. *Both sides* (past and future references) combined with Poisson weights: worse than taking the side with more elements.

- *Chained detours* through related edges and short overlap estimates: much worse ($\Delta\mathcal{C} = -0.06$ to -0.29); the short legs are biased low and the minimum over many routes selects the lowest.

- *Cheaper detours* through random intermediate elements: at or below Johnston (-0.008 to -0.033 at $n = 3000$). Through intermediate elements chosen near the midpoint of each distant pair on a rough first map: about two thirds of the switch's gain at $n = 3000$ at a quarter of the cost, but still cubic.

[1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, Spacetime as a causal set, Phys. Rev. Lett. **59**, 521 (1987).
[2] S. Surya, The causal set approach to quantum gravity, Living Rev. Relativ. **22**, 5 (2019), arXiv:1903.11544.
[3] G. Brightwell and R. Gregory, Structure of random discrete spacetime, Phys. Rev. Lett. **66**, 260 (1991).

[4] L. Glaser and S. Surya, Towards a definition of locality in a manifoldlike causal set, Phys. Rev. D **88**, 124026 (2013), arXiv:1309.3403.

[5] M. Roy, D. Sinha, and S. Surya, Discrete geometry of a small causal diamond, Phys. Rev. D **87**, 044046 (2013), arXiv:1212.0631.

[6] D. M. T. Benincasa and F. Dowker, The scalar curvature of a causal set, Phys. Rev. Lett. **104**, 181301 (2010), arXiv:1001.2725.

[7] D. Rideout and P. Wallden, Spacelike distance from discrete causal order, Classical Quantum Gravity **26**, 155013 (2009), arXiv:0810.1768.

[8] M. Boguñá and D. Krioukov, Measuring spatial distances in causal sets via causal overlaps, Phys. Rev. D **110**, 024008 (2024), arXiv:2401.17376.

[9] C. A. Hu, D. A. Meyer, and E. J. Q. Meyer, Reconstructing Minkowski geometry from causal separations, J. Math. Phys. **66**, 122502 (2025), arXiv:2601.03280.

[10] J. Henson, Constructing an interval of Minkowski space from a causal set, Classical Quantum Gravity **23**, L29 (2006), arXiv:gr-qc/0601069.

[11] J. R. Clough and T. S. Evans, Embedding graphs in Lorentzian spacetime, PLoS One **12**, e0187301 (2017), arXiv:1602.03103.

[12] S. Johnston, Embedding causal sets into Minkowski spacetime, Classical Quantum Gravity **39**, 095006 (2022), arXiv:2111.09331.

[13] S. Johnston, Simpler embeddings of causal sets into Minkowski spacetime, Phys. Rev. D **111**, 106020 (2025), arXiv:2502.09701.

[14] G. Casasola, Causal sets that locate and grow themselves: sub-discreteness self-localization and order-intrinsic sequential growth of 3+1 Minkowski orders, Research Square preprint (2026), doi:10.21203/rs.3.rs-11223018/v1.

[15] R. Bada, Counting identities in two-dimensional causal sets: spatial distance, depth-drift curvature, and a fixed-size propagator target (companion paper, 2026), to be submitted.

[16] G. Pellegrin, Link-restricted path filling for causal set embeddings, Zenodo preprint (2026), doi:10.5281/zenodo.21813653.

[17] O. Müller, On the Hauptvermutung of causal set theory, arXiv:2503.01719 (2025).

[18] C. Buschmann, D. Pfisterer, and S. Fischer, Estimating distances using neighborhood intersection, in *Proceedings of the IEEE Conference on Emerging Technologies and Factory Automation (ETFA)* (IEEE, 2006).

[19] B. Huang, C. Yu, B. D. O. Anderson, and G. Mao, Estimating distances via connectivity in wireless sensor networks, Wirel. Commun. Mob. Comput. **14**, 541 (2014).

[20] J. de Leeuw and P. Mair, Multidimensional scaling using majorization: SMACOF in R, J. Stat. Softw. **31**(3), 1 (2009).

[21] J. A. Costa, N. Patwari, and A. O. Hero, Distributed weighted-multidimensional scaling for node localization in sensor networks, ACM Trans. Sens. Netw. **2**, 39 (2006).

[22] J. O. Ramsay, Maximum likelihood estimation in multidimensional scaling, Psychometrika **42**, 241 (1977).

[23] P. D. Hoff, A. E. Raftery, and M. S. Handcock, Latent space approaches to social network analysis, J. Am. Stat. Assoc. **97**, 1090 (2002).

[24] M. Boguñá, F. Papadopoulos, and D. Krioukov, Sustaining the Internet with hyperbolic mapping, Nat. Commun. **1**, 62 (2010), arXiv:1009.0267.

[25] G. García-Pérez, A. Allard, M. Á. Serrano, and M. Boguñá, Mercator: uncovering faithful hyperbolic embeddings of complex networks, New J. Phys. **21**, 123033 (2019), arXiv:1904.10814.

[26] A. Sim, M. Wiatrak, A. Brayne, P. Creed, and S. Paliwal, Directed graph embeddings in pseudo-Riemannian manifolds, in *Proceedings of ICML 2021*, arXiv:2106.08678.

[27] W. J. Cunningham, Inference of boundaries in causal sets, Classical Quantum Gravity **35**, 094002 (2018), arXiv:1710.09705.

[28] N. Madsen, Reconstruction of the Riemann tensor from causal order and element counts, arXiv:2610.36695 (2026).

[29] R. Bada, Two causal-set papers written with an AI research partner: methodology note (2026), https://github.com/presindev/causet-counting.